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MIT SCHOOL OF COMPUTING

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DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

Ledgeit - A manufacturing platform that records AI-generated alerts, stores evidence hashes on blockchain, produces tamper-proof audit trails, displays event timelines and supports compliance reporting

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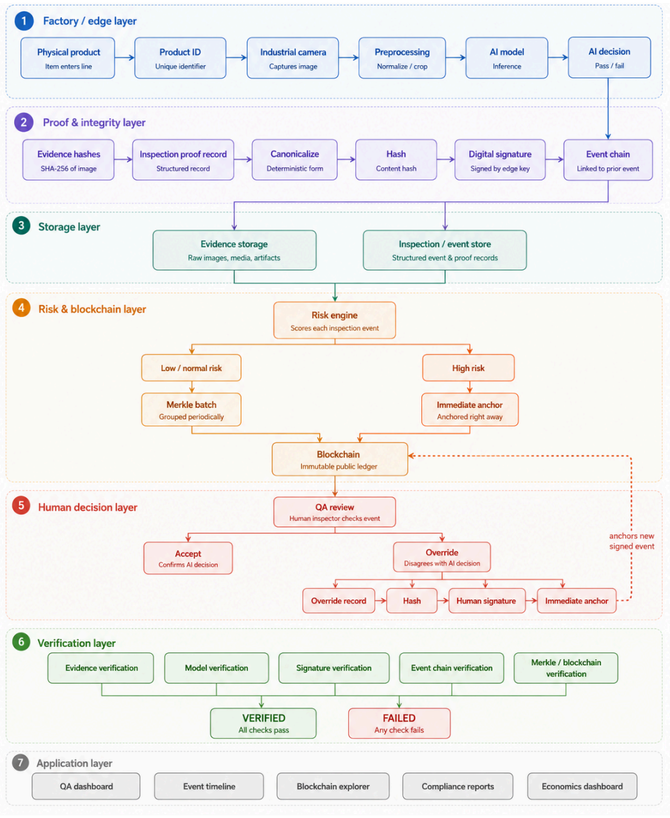
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Problem Statement : Factories now let AI inspect every product for defects — but if something later goes wrong, there's no way to prove what the AI actually saw, decided, or whether that record was quietly changed afterward. We've seen this exact failure before: manufacturers editing inspection data to hide problems. Our platform locks the AI's evidence, decision, and any human override together the moment it happens — so nobody, not even us, can rewrite what really occurred.

PROPOSED SOLUTION

- **Stage 1** — **Detect** An AI camera system inspects every product as it comes off the line, spotting defects — cracks, dents, bad welds — and recording its decision along with how confident it is..
- **Stage 2** — **Bind** The moment a decision is made, we lock together everything that matters: the exact image, which AI version made the call, its reasoning, and the confidence score — as one sealed unit.
- **Stage 3** — **Anchor** That sealed unit gets a unique digital fingerprint, recorded on a public, tamper-evident ledger. Routine checks are grouped together to save cost; serious defects get recorded immediately, without delay.
- **Stage 4** — **Respond** If a human operator disagrees with the AI and overrides it, that decision — who, why, and what evidence they reviewed — gets sealed and recorded the same way, permanently.
- **Stage 5** — **Verify** Anytime later, anyone can re-check a record against its original fingerprint. If even one detail was altered, the mismatch is caught and flagged immediately — no exceptions, no cover-ups.
- **Stage 6** — **Report** All verified records automatically compile into audit-ready reports for inspectors, regulators, or customers — proof the factory's quality process actually happened the way it claims it did.

BLOCK DIAGRAM



CASE STUDIES WHICH EXPLAINS THE PROBLEM

- **Kobe Steel (2017–18)** — Fabricated inspection data across **23** plants, affecting **200+** companies and **20,000+** tons of materials.
- **Mitsubishi Materials** — Admitted to rewriting inspection records, closely matching post-inspection data **tampering**.
- **Global Calcium (FDA, 2024)** — **Fabricated** batch records and deleted **evidence** before inspection, showing deliberate record manipulation and deletion.
- **Nidec (2026)** — **1,000+** alleged unauthorized quality/design changes, showing quality **fraud** is still an active industry problem.

SCOPE AND FEASIBILITY

- **Integration-Friendly** — Works as an integrity layer over existing AI inspection and factory systems, so **manufacturers don't need to replace their current infrastructure**.
- **Scalable & Cost-Feasible** — Heavy evidence stays off-chain; routine records are batch-anchored while only critical events are anchored immediately.
- **Independent Auditability** — Auditors can **independently verify evidence**, AI decisions, model versions, and human overrides instead of relying solely on editable internal records.